Detailed derivation of the best frequency for inverse-bremsstrahlung heating of krypton laser gases, with a worked example

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These notes derive, with every intermediate result written out, the frequency at which an electromagnetic wave heats the free electrons of a krypton laser gas best by inverse bremsstrahlung, for two gases, pure krypton and krypton with 2 % fluorine, at electron densities of 10^{14} *to* 10^{16} cm^{-3}*, pressures of 0.1 to 2 atm, and 100 to 500* kW/cm^3 *deposited for a microsecond. The absorption coefficient is the Drude form with a collision frequency built from the momentum-transfer cross sections of the atoms through the two-term kinetic equation, and from the exact free–free Gaunt factor for the ions. The electron temperature follows from the power balance, the best frequency from the depth of the gas, the choice between the GHz and the THz ranges from the absorption across the spectrum, and the uncertainty of the result from the envelope against a kinetic solution, from the electron–ion term against the measurements of Turnbull et al., and from the cross sections. A worked example at one operating point carries the whole chain to a recommended frequency. Every boxed result and every printed number is checked numerically by* `envelope_v6_check.py`*, which Appendix A describes.*

For notation, Gaussian units are used in the derivations, as in [1], and practical units in the numbers. The electron has charge $-e$ and mass m. The atoms have density N and fractions x_g, the electrons density n_e and temperature T_e, the ions density $n_i = n_e$ and charge e. The wave has angular frequency $\omega = 2\pi f$ and the field $E_0 \cos \omega t$, linearly polarized. The thermal velocity is $v_t = (k_B T_e/m)^{1/2}$ and the plasma frequency $\omega_p = (4\pi n_e e^2/m)^{1/2}$. An electron energy is $\varepsilon = mv^2/2$. A boxed equation is a result that is used later. An equation of another document is cited by its own number followed by the reference, as in Eq. (11.9) [1], and an equation number without a reference is one of these notes.

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1 The question and what best means

(1) A wave is to deposit a power P of 100 to 500 kW/cm^3 for a microsecond into the free electrons of a laser gas, of density $n_e = 10^{14}$ to 10^{16} cm^{-3}, at $p = 0.1$ to 2 atm and 300 K, where the gas density is $N = 2.45 \times 10^{19}\, p$ atoms per cm^3 per atmosphere. The electrons absorb the wave by inverse bremsstrahlung, while they collide with the atoms and, far less often, with the ions. Which frequency does this best?

(2) A wave below the plasma frequency is reflected. A wave just above it is absorbed in a thin sheet at the front of the gas. A wave far above it crosses the gas and leaves. The best frequency is therefore the one whose absorption length matches the depth L of the gas, provided that the gas reflects little of it:

$$\boxed{f_{\text{best}}(L) = \max\big(f_A, f_R\big), \qquad \alpha(f_A)\, L = 1, \qquad R(f_R) = 5\,\%,} \tag{1.1}$$

with α the absorption coefficient of the intensity and R the reflection of the front surface at normal incidence, f_R the lowest frequency that reflects no more than 5 %. At f_A one pass absorbs $1 - e^{-1} = 63\,\%$ of what enters and the deposition at the back is e^{-1} of that at the front. Asking for $\alpha L = q$ instead moves f_A by $q^{-1/2}$ where $\alpha \propto f^{-2}$. The reference depth is $L = 10$ cm, and Sec. 4 shows $f_A \propto L^{1/2}$.

(3) Two gases are compared: pure Kr, and Kr with 2 % F_2. The momentum-transfer and

inelastic cross sections are the Biagi set for krypton [2] and Morgan's set for F_2 [3].

(4) The plan. Section 2 derives the absorption coefficient and the collision frequency of the atoms and of the ions. Section 3 derives the electron temperature from the power balance, since what the atoms scatter depends on it. Section 4 derives the best frequency in closed form. Section 5 follows the absorption across the spectrum and settles the GHz range against the THz range. Section 6 gives the results for the two gases with the particle balance. Section 7 checks the Maxwellian envelope against a two-term kinetic solution, Sec. 8 derives the uncertainty of every entry, and Sec. 9 works one operating point through from the gas density to the recommended frequency.

2 The absorption coefficient

2.1 The Drude model

This subsection follows [1, Sec. 11.1.1], in whose convention a perturbation varies as $e^{-i\omega t}$ and the physical quantity is the real part.

(1) The electron velocity obeys Newton's equation with a friction force that stands for the collisions,

$$-i\omega m\mathbf{v} = -e\mathbf{E} - \nu m\mathbf{v} \quad \longrightarrow \quad \mathbf{v} = -\frac{e\mathbf{E}}{m(\nu - i\omega)} = -\frac{ie\mathbf{E}}{m(\omega + i\nu)}, \tag{2.1}$$

which is Eq. (11.7) [1], with $\nu - i\omega = -i(\omega + i\nu)$ in the last step.

(2) The current density and the conductivity are

$$\mathbf{j} = -en_e\mathbf{v} = \frac{ie^2 n_e}{m(\omega + i\nu)}\,\mathbf{E}, \qquad \sigma = \frac{ie^2 n_e}{m(\omega + i\nu)} = \frac{e^2 n_e}{m(\nu - i\omega)}, \tag{2.2}$$

and with the dielectric function $\varepsilon = 1 + (4\pi i/\omega)\sigma$ of Eq. (11.4) [1],

$$\boxed{\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega(\omega + i\nu)} = 1 - \frac{\omega_p^2}{\omega^2 + \nu^2} + i\,\frac{\omega_p^2\,\nu}{\omega(\omega^2 + \nu^2)}.} \tag{2.3}$$

(3) A plane wave $e^{i(kz-\omega t)}$ has $k = (\omega/c)\sqrt{\varepsilon}$. With $\sqrt{\varepsilon} = n_r + i\kappa$ the intensity falls as $e^{-2\,\mathrm{Im}\,k\,z}$, and the reflection of a sharp boundary at normal incidence is that of the Fresnel formula,

$$\boxed{\alpha = \frac{2\omega}{c}\,\kappa, \qquad R = \left|\frac{1 - \sqrt{\varepsilon}}{1 + \sqrt{\varepsilon}}\right|^2.} \tag{2.4}$$

Where the absorption is weak, $\mathrm{Im}\,\varepsilon \ll \mathrm{Re}\,\varepsilon$, one has $2\kappa \simeq \mathrm{Im}\,\varepsilon/n_r$ with $n_r \simeq (\mathrm{Re}\,\varepsilon)^{1/2}$, so that

$$\boxed{\alpha = \frac{\omega}{c}\,\frac{\mathrm{Im}\,\varepsilon}{(\mathrm{Re}\,\varepsilon)^{1/2}} = \frac{\nu}{c}\,\frac{\omega_p^2}{\omega^2}\Big(1 - \frac{\omega_p^2}{\omega^2}\Big)^{-1/2}\Big[1 + O\Big(\frac{\nu^2}{\omega^2}\Big)\Big],} \tag{2.5}$$

which is Eq. (11.12) [1]. Equation (2.4) is used in the numbers and Eq. (2.5) in the closed forms.

(4) The power absorbed per unit volume is the cycle average of $\mathbf{j}\cdot\mathbf{E}$,

$$\boxed{P = \tfrac{1}{2}\,\mathrm{Re}\,\sigma\,E_0^2 = \frac{n_e e^2}{2m}\,\frac{\nu}{\omega^2+\nu^2}\,E_0^2.} \tag{2.6}$$

Everything that distinguishes one model of inverse bremsstrahlung from another sits in ν, which the next two subsections build from the collisions with the atoms and with the ions.

2.2 The collision frequency of the atoms

(1) In a weakly ionized gas the electrons collide with atoms. The momentum-transfer collision frequency of an electron of speed v is $\nu_m(v) = N\sigma_m(v)v$, with σ_m the momentum-transfer cross section of the mixture, $\sigma_m = \sum_g x_g\sigma_{m,g}$. It depends on v, strongly so near a Ramsauer minimum, and the single ν of Eq. (2.1) has to be an average of it with the right weight.

(2) The weight follows from the two-term kinetic equation [4]. Write the velocity distribution as $f_0(v) + f_1(v)\cos\theta$, with θ the angle to the field and $\int 4\pi v^2 f_0\,\mathrm{d}v = n_e$. The anisotropic part obeys, for the field $\mathrm{Re}[E_0\hat{\mathbf{e}}e^{-i\omega t}]$,

$$-i\omega f_1 + \nu_m(v)\,f_1 = \frac{eE_0}{m}\,\frac{\partial f_0}{\partial v} \quad\longrightarrow\quad f_1 = \frac{eE_0}{m}\,\frac{\partial_v f_0}{\nu_m(v) - i\omega}, \tag{2.7}$$

where the collisions relax f_1 at the rate ν_m and leave f_0 alone.

(3) The current is the first moment of f_1,

$$j = -e\int v\cos\theta\, f_1\cos\theta\,\mathrm{d}^3v = -\frac{4\pi e}{3}\int_0^\infty v^3 f_1\,\mathrm{d}v = -\frac{4\pi e^2 E_0}{3m}\int_0^\infty \frac{v^3\,\partial_v f_0}{\nu_m(v) - i\omega}\,\mathrm{d}v, \tag{2.8}$$

and its real part, which carries the absorption, is

$$\mathrm{Re}\,\sigma = -\frac{4\pi e^2}{3m}\int_0^\infty \frac{v^3\nu_m(v)}{\nu_m^2(v)+\omega^2}\,\partial_v f_0\,\mathrm{d}v. \tag{2.9}$$

(4) Compare with the Drude form, $\mathrm{Re}\,\sigma = n_e e^2\nu/[m(\nu^2+\omega^2)]$ from Eq. (2.2). For $\omega \gg \nu_m$ both denominators are ω^2 and the comparison defines the effective collision frequency of the atoms,

$$\nu_{ea} = -\frac{4\pi}{3n_e}\int_0^\infty v^3\nu_m(v)\,\partial_v f_0\,\mathrm{d}v. \tag{2.10}$$

For a Maxwellian, $f_0 \propto e^{-v^2/2v_t^2}$ and $\partial_v f_0 = -(v/v_t^2)f_0$, so that

$$\boxed{\nu_{ea} = \frac{\langle\nu_m v^2\rangle}{3v_t^2} = N\Big\langle\sigma_m v\,\frac{2\varepsilon}{3k_B T_e}\Big\rangle,} \tag{2.11}$$

the average over the Maxwellian with the energy as a weight: the field heats the fast electrons most. For a constant σ_m this is 4/3 of the Drude mean $\langle\nu_m\rangle$, and for the two gases of these notes it is 1.72 to 2.23 times $\langle\nu_m\rangle$ at the cells of Table 2 where $T_e \geq 0.5$ eV, since σ_m of krypton climbs steeply above its Ramsauer minimum at 0.5 eV. By the low-energy theorem [5, 6] this kernel is exact for photons of energy small against the electron's, and the atom enters only through σ_m.

(5) The same collisions take energy from the electrons. An electron of energy ε that scatters elastically through the angle χ on an atom of mass $M \gg m$ at rest loses $2(m/M)\varepsilon(1-\cos\chi)$, and the average of $1-\cos\chi$ over the angular distribution defines σ_m, so that the loss rate is $(2m/M)\nu_m(v)\varepsilon$ per electron. Over the Maxwellian, with the gas at T_g giving back k_BT_g for k_BT_e,

$$\boxed{P_{\rm el} = \sum_g x_g \,\frac{2m}{M_g}\, N\langle \sigma_{m,g} v\, \varepsilon\rangle \Big(1 - \frac{T_g}{T_e}\Big).} \tag{2.12}$$

2.3 The collision frequency of the ions

(1) For electrons on ions of charge e the Drude collision frequency is $\nu_{ei} = \nu_T \Lambda$ with the Spitzer frequency

$$\boxed{\nu_T = \frac{4\sqrt{2\pi}}{3}\,\frac{n_i e^4}{m^{1/2}(k_BT_e)^{3/2}} = 2.91\times10^{-12}\,\frac{n_i\,[\mathrm{m}^{-3}]}{(k_BT_e\,[\mathrm{eV}])^{3/2}}\ \mathrm{s}^{-1},} \tag{2.13}$$

which is Eq. (10.133) [1] without its logarithm, and Λ holds the physics of the Coulomb collision.

(2) The quantum theory of inverse bremsstrahlung gives Λ as a Gaunt factor. The absorption, net of stimulated emission, of a Maxwellian is

$$\boxed{\Lambda = \frac{\pi}{\sqrt{3}}\,\langle g\rangle(\gamma^2, u)\,\frac{1-e^{-u}}{u}, \qquad u = \frac{\hbar\omega}{k_BT_e}, \quad \gamma^2 = \frac{\mathrm{Ry}}{k_BT_e},} \tag{2.14}$$

with $\langle g\rangle$ the thermally averaged free–free Gaunt factor of the exact Coulomb wave functions [7], tabulated by van Hoof et al. [8] against γ^2 and u. Its Born limit, $\gamma^2 \to 0$, is

$$\Lambda_1(u) = \frac{1-e^{-u}}{u}\,e^{u/2}K_0\Big(\frac{u}{2}\Big) \simeq \ln\frac{4k_BT_e}{e^\gamma\hbar\omega} = \ln\frac{b_{\max}}{b_{\min}}, \qquad b_{\max} = \frac{v_t}{\omega}, \quad b_{\min} = \frac{e^\gamma}{4}\,\frac{\hbar}{mv_t}, \tag{2.15}$$

for $u \ll 1$, with γ Euler's constant in the exponent: the upper cutoff is the distance an electron travels in a radian of the period, the dynamic cutoff of Dawson and Oberman [9], which Turnbull et al. measured to be the right one [10], and not the Debye length.

(3) The Born limit needs $\eta_t = e^2/\hbar v_t \ll 1$. Here k_BT_e is about an electron volt and $\eta_t = 3.2$ to 5.3 at the warm cells: the collisions are classical, and in the soft-photon limit the exact logarithm of one electron is the Born one plus Bloch's term $-\gamma - \mathrm{Re}\,\psi(1+i\eta)$, which for $\eta \gg 1$ is $-\gamma - \ln\eta$ and turns the lower cutoff into the classical $e^\gamma e^2/2mv^2$. The tabulated $\langle g\rangle$ contains this: at the warm cells $\Lambda = 2.5$ to 5.6, where the Born logarithm Λ_1 would give 3.0 to 7.1.

(4) The two kinds of scatterer add, since electron–ion and electron–atom collisions are separate events and electron–electron collisions radiate no dipole:

$$\boxed{\nu_{\rm eff} = \nu_{ea} + \nu_T\Lambda.} \tag{2.16}$$

At the cells of Table 2 with $T_e \geq 0.5$ eV, $\nu_{\rm eff} = 2.7\times10^{11}$ to 5.7×10^{12} s^{-1}, and the ions' share of it is below 2 % at 10^{15} cm^{-3} and 3 to 26 % at 10^{16} cm^{-3}, the most at 0.1 atm.

2.4 What is left out, and its size

(1) Screening by the electrons and the correlations of the ions change Λ by a few per cent at most at these densities. (2) The quantum reduction of the atoms' term at a finite photon energy is below 1 % for $u \le 0.12$, which holds at every warm cell. (3) Langdon's flattening of the distribution by the ion heating has the parameter $\alpha_L = v_{\rm os}^2/v_t^2$ with the quiver velocity $v_{\rm os} = eE_0/m\omega$; from Eq. (2.6) at $\omega \gg \nu$,

$$\boxed{\alpha_L = \frac{v_{\rm os}^2}{v_t^2} = \frac{e^2E_0^2}{m^2\omega^2 v_t^2} = \frac{2\,P/n_e}{\nu_{\rm eff}\,k_BT_e},} \tag{2.17}$$

where the factor $1+\nu^2/\omega^2$ left out is below 1.06 at the warm cells. It is at most 3×10^{-3} in Table 2 wherever the ions carry more than 2 % of the absorption, and Langdon's factor [12] then changes the absorption by less than 1 % at the warm cells. (4) The reflection is that of a sharp boundary, an upper bound for a graded one.

In Summary: The absorption coefficient is the Drude form (2.4) with the collision frequency (2.16): the atoms through the energy-weighted average (2.11) of their momentum-transfer collision frequency, exact for soft photons, and the ions through the Spitzer frequency (2.13) and the exact Gaunt factor (2.14), whose cutoffs are the dynamic one and the classical one. Both depend on T_e, which the next section fixes.

3 The electron temperature

(1) The field delivers P/n_e to each electron, and in a steady state the electrons lose it by elastic recoil, Eq. (2.12), and by the inelastic collisions with threshold energies ε_j:

$$\boxed{\frac{P}{n_e} = P_{\rm el}(T_e) + N\sum_j x_j k_j(T_e)\,\varepsilon_j,} \tag{3.1}$$

which fixes T_e for a Maxwellian. The electrons that attachment removes leave with their own energy and are replaced by the source that holds the density, so attachment is not a loss in Eq. (3.1). Per electron, P/n_e runs from 6.2×10^7 eV/s at 10^{16} cm^{-3} and 100 kW/cm^3 to 3.1×10^{10} eV/s at 10^{14} cm^{-3} and 500 kW/cm^3.

(2) The rate coefficient of a process of cross section $\sigma_j(\varepsilon)$ is $k_j = \langle\sigma_j v\rangle$. With the Maxwellian written in the energy, $F(\varepsilon) = \frac{2}{\sqrt{\pi}}(k_BT_e)^{-3/2}e^{-\varepsilon/k_BT_e}$ normalized by $\int_0^\infty \varepsilon^{1/2}F\,{\rm d}\varepsilon = 1$, and $v = \gamma_e\varepsilon^{1/2}$ with $\gamma_e = (2/m)^{1/2}$,

$$\boxed{k_j = \int_0^\infty \sigma_j(\varepsilon)\,v\,\varepsilon^{1/2}F(\varepsilon)\,{\rm d}\varepsilon = \gamma_e\int_0^\infty \varepsilon\,\sigma_j(\varepsilon)\,F(\varepsilon)\,{\rm d}\varepsilon,} \tag{3.2}$$

and the same weight εF gives Eq. (2.11) as $\nu_{ea} = N\gamma_e\int\varepsilon\sigma_m(2\varepsilon/3k_BT_e)F\,{\rm d}\varepsilon$ and Eq. (2.12) with $\varepsilon^2\sigma_m F$.

(3) In pure Kr the recoil takes 0.1 to 24 % of the power and the 5s excitation of krypton at 9.9 eV the rest, so T_e sits where the Maxwellian tail reaches 10 eV: 0.95 to 2.6 eV at 300 kW/cm^3. With 2 % F_2 the vibrational channels (0.1 to 0.4 eV) and the dissociative ones (3.2 and 4.3 eV) take 1 to 13 % of the losses at 10^{14} cm^{-3}, 8 to 74 % at 10^{15} and 45 to 100 % at 10^{16},

where the power per electron is least, and lower T_e to 1.05 eV at 10^{15} cm^{-3} and 2 atm and, at 10^{16} cm^{-3}, to 1.23 eV at 0.1 atm and to 0.19 and 0.088 eV at 0.5 and 2 atm.

(4) At the two cells where F_2 holds T_e at 0.19 and 0.088 eV, in and below the Ramsauer minimum of krypton, the ions carry 79 to 92 % of $\nu_{\rm eff}$, u reaches 0.30 and 1.0, outside the soft-photon range of the atoms' term, and no source could hold the electrons (Sec. 6.2). Their entries in the tables are formal.

(5) The field itself follows from Eq. (2.6) once $\alpha L = 1$ ties the frequency to the collision frequency. With Eq. (2.5), $\nu/(\omega^2+\nu^2) = c\,n_r/(\omega_p^2 L)$ at f_A, and Eq. (2.6) becomes

$$\boxed{E_0 = \Big(\frac{8\pi P L}{c\,n_r}\Big)^{1/2} = 47.5\ \text{kV/cm}\,\Big(\frac{P}{300\ \text{kW/cm}^3}\,\frac{L}{10\ \text{cm}}\Big)^{1/2} n_r^{-1/2},} \tag{3.3}$$

the same at every cell but for n_r, since the absorbed power per volume is the intensity over the absorption length: the intensity is PL in that sense, 3 MW/cm^2 at 300 kW/cm^3 and 10 cm.

In Summary: The power balance (3.1) with the Maxwellian rates (3.2) sets T_e at 0.95 to 2.6 eV at the warm cells of the two gases. The 5s level of krypton takes the power in pure Kr, and F_2's low channels take over where the power per electron is least and cool the electrons into the Ramsauer minimum.

4 The best frequency

4.1 The reflection limit

(1) For $\nu \ll \omega$ the dielectric function (2.3) is real, $\varepsilon = 1 - \omega_p^2/\omega^2$, and above the cutoff $\sqrt{\varepsilon} = n_r$ is real too, so that Eq. (2.4) gives $R = [(1-n_r)/(1+n_r)]^2$. Solving for $R = 5\,\%$,

$$n_r = \frac{1-\sqrt{R}}{1+\sqrt{R}} = 0.6345, \qquad \boxed{\frac{f_R}{f_{pe}} = \big(1-n_r^2\big)^{-1/2} = 1.294 \quad (\nu \ll \omega_p),} \tag{4.1}$$

with the cutoff $f_{pe} = \omega_p/2\pi = 8.98\ \text{kHz}\,(n_e/\text{cm}^{-3})^{1/2}$, which is 0.090, 0.28 and 0.90 THz at the three densities. So $f_R \le 1.16$ THz, at 10^{16} cm^{-3}. Where ν exceeds ω_p, at 10^{14} cm^{-3} from 0.5 atm up, the collisions damp the reflection and f_R falls to 0.03 THz and below, far under f_A.

4.2 The absorption length

(1) Well above the cutoff and the collision frequency Eq. (2.5) is $\alpha = \omega_p^2 \nu_{\rm eff}/(c\,\omega^2)$, and $\alpha L = 1$ gives

$$\boxed{f_A = \frac{1}{2\pi}\Big(\frac{\omega_p^2\,\nu_{\rm eff}\,L}{c}\Big)^{1/2} = 5.2\ \text{THz}\Big(\frac{n_e}{10^{15}\ \text{cm}^{-3}}\,\frac{\nu_{\rm eff}}{10^{12}\ \text{s}^{-1}}\,\frac{L}{10\ \text{cm}}\Big)^{1/2},} \tag{4.2}$$

within 3.1 % of the exact root of $\alpha L = 1$ with Eq. (2.4) at every one of the 54 points computed below. At every one of them $f_A > f_R$: the absorption length, not the cutoff, sets the frequency.
(2) Since $\nu_{\rm eff} \simeq \nu_{ea} \propto N\sigma_m(T_e)$, the best frequency grows as the square root of the electron density, of the gas density, and of the depth. The exponents come out below 1/2 in the fits of Sec. 6.3, because the hotter, thinner cells scatter more per atom: T_e falls with $n_e p$ through Eq. (3.1), and σ_m with it.

In Summary: The best frequency is Eq. (4.2), the frequency whose absorption length is the depth of the gas, bounded below by the reflection limit (4.1), which never binds here.

5 The spectrum: GHz or THz?

(1) Before the best frequency is read off, the whole spectrum is worth a look, since the microwave sources sit in the GHz range. With each cell's $\nu_{\rm eff}$ and ω_p, Eqs. (2.3) and (2.4) give $\alpha(f)$ and $R(f)$ from 1 GHz to 100 THz, the ions' logarithm taken at each frequency. Three regimes appear. Below the collision frequency the wave enters as a collisional skin: with $\omega \ll \nu$ and $\omega_p^2 \gg \omega\nu$, Eq. (2.3) is dominated by its imaginary part, $\varepsilon \simeq i\omega_p^2/\omega\nu$, so that $\sqrt{\varepsilon} \simeq (1+i)(\omega_p^2/2\omega\nu)^{1/2}$ and

$$\boxed{\alpha \simeq \frac{\omega_p}{c}\Big(\frac{2\omega}{\nu_{\rm eff}}\Big)^{1/2} \qquad (\omega \ll \nu_{\rm eff}),} \tag{5.1}$$

rising as $f^{1/2}$: at 2.45 GHz this law reproduces the exact α within a factor 0.98 to 1.14 at the 18 cells. Near the larger of the collision frequency and the cutoff, α peaks, at 0.05 to 0.50 THz over the 18 cells, and the absorption length is shortest there, 0.03 to 1.7 mm for pure Kr at 0.5 atm. Above both, the gas is transparent, α falls as f^{-2} by Eq. (2.5), and $R \to 0$. Here $\nu_{\rm eff}/2\pi = 0.009$ to 0.91 THz and $f_{pe} = 0.090$ to 0.90 THz, so the boundary between the two bands is where the physics turns: the GHz band lies on the skin side and the THz band on the transparent side.

(2) Table 1 and Fig. 1 give pure Kr at 0.5 atm: the absorption length $1/\alpha$, the reflection R, and the fraction $A = (1-R)(1-e^{-\alpha L})$ of the incident power deposited in the 10 cm.

Table 1: Pure Kr at 0.5 atm, from 2.45 GHz to 30 THz: the absorption length, the reflection in per cent, and the fraction of the incident power deposited in $L = 10$ cm, in per cent, at the three densities.

	10^{14} cm^{-3}			10^{15} cm^{-3}			10^{16} cm^{-3}		
f	$1/\alpha$	R	A	$1/\alpha$	R	A	$1/\alpha$	R	A
2.45 GHz	4.2 mm	43	57	1.1 mm	80	20	0.31 mm	94	6
10 GHz	2.4 mm	17	83	0.52 mm	65	35	0.15 mm	88	12
30 GHz	1.8 mm	4	96	0.30 mm	49	51	0.08 mm	82	18
95 GHz	1.8 mm	1	99	0.17 mm	31	69	0.04 mm	75	25
300 GHz	3.5 mm	0	100	0.26 mm	7	93	0.03 mm	70	30
1 THz	2.2 cm	0	99	3.0 mm	0	100	0.19 mm	13	87
3 THz	18.7 cm	0	41	2.7 cm	0	97	3.5 mm	0	100
10 THz	2 m	0	5	30.5 cm	0	28	4.2 cm	0	91
30 THz	19 m	0	1	3 m	0	4	38.1 cm	0	23

(3) *GHz.* At 10^{15} and 10^{16} cm^{-3} the gas is a mirror: it reflects 31 to 94 % between 2.45 and 95 GHz at 0.5 atm, 8 to 96 % over all pressures, and absorbs the remainder within 0.04 to 1.1 mm of the front. At 10^{14} cm^{-3} the collisions outrun the cutoff and the wave gets in (43 % reflected at 2.45 GHz and 1 % at 95 GHz), but it is absorbed in 1.8 to 4.2 mm (0.6 to 8.3 mm over the pressures and both gases): the front millimetres take all of it and the rest of the gas nothing, since the deposition falls by $e^{\alpha L}$, at least 10^5, from front to back.

THz. The gas reflects at most 15 % (1 THz at 10^{16} cm^{-3}) and nothing above 3 THz, and the absorption length runs from 0.2 mm to 2 m, through the depth of the gas at 2.2, 5.7 and 15.4 THz for the three densities at 0.5 atm, and at 1.1 to 26.5 THz over the 18 cells. At 3 THz the 10 cm deposits 41, 97 and 100 % of the incident power. Per electron the heating at a given intensity goes as $\nu_{\rm eff}/(\omega^2 + \nu_{\rm eff}^2)$ by Eq. (2.6), largest at GHz, where every electron heats as in a

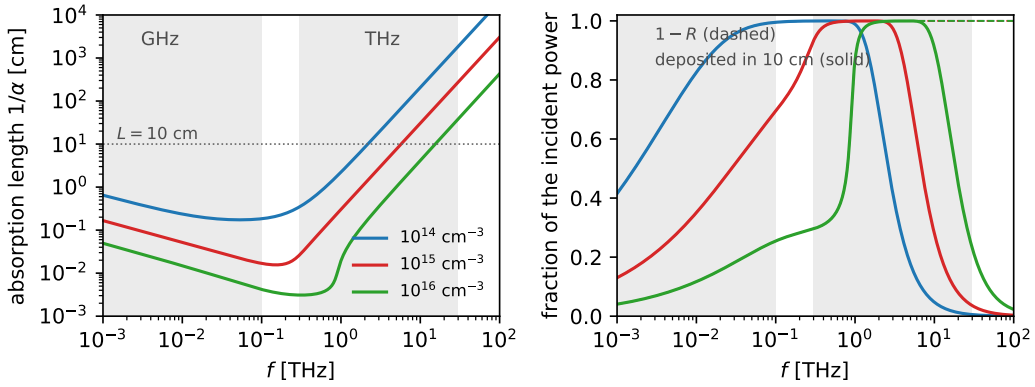


Figure 1: Pure Kr at 0.5 atm. Left: the absorption length against frequency for the three densities, with the depth $L = 10$ cm dotted; the shaded bands are the GHz range (1 to 100 GHz) and the THz range (0.3 to 30 THz). Right: the fraction of the incident power that enters the gas, $1 - R$ (dashed), and the fraction deposited in the 10 cm (solid).

static field, and that is exactly why the skin is thin: the first electrons take the whole wave. The GHz range would be the right one only for a gas a millimetre thick at 10^{14} cm^{-3} and 0.5 atm or more, where 95 GHz enters with 1 % reflection and is absorbed in 1.8 to 5.4 mm, or at densities below 10^{12} cm^{-3}, where the cutoff falls to 9 GHz. Kr/F_2 changes the absorption lengths by at most a factor 1.3 at 95 GHz and 3 THz and the reflections by at most 9 points.

In Summary: For a gas deeper than a few millimetres the THz range is better: it enters the gas and deposits over centimetres, where the GHz range reflects or skins. The best frequency of Eq. (4.2) lies on the transparent side of the peak, in the THz range at every cell.

6 The two gases

6.1 The best frequencies

Table 2 gives f_{best} at $L = 10$ cm and 300 kW/cm^3 for the two gases, with T_e from Eq. (3.1). The flags come from the particle balance of Sec. 6.2.

6.2 The particle balance

(1) The density is not the field's to choose. Direct ionization by the electrons, at the rate $\nu_{\text{ion}} = N \sum_j x_j k_{\text{ion},j}$ from the thresholds at 14.0 eV (Kr) and 15.7 eV (F_2), makes electrons. Attachment to F_2 at $\nu_{\text{att}} = N x_{F_2} k_{\text{att}}$ removes them, and so does recombination. For the recombination every cation is taken as Kr_2^+ with the coefficient of Shiu and Biondi [16],

$$\alpha_{\text{rec}} = 1.6 \times 10^{-6} \Big(\frac{300\ \text{K}}{T_e} \Big)^{0.55}\ \text{cm}^3/\text{s}, \qquad \nu_{\text{rec}} = \alpha_{\text{rec}}\, n_e, \tag{6.1}$$

measured from 300 to 8400 K, 0.72 eV, and extrapolated here. The net rate is $\nu_{\text{net}} = \nu_{\text{ion}} - \nu_{\text{att}} - \nu_{\text{rec}}$.

(2) Where $\nu_{\text{net}} > 0$ the density e-folds in $1/\nu_{\text{net}}$, which is 2.4 to 5.2 ns at the four cells

Table 2: The best frequency in THz at $L = 10$ cm and 300 kW/cm^3, with T_e in eV in parentheses, for the two gases. ^{a}: the field's own ionization outruns the losses and n_e grows during the pulse. ^{s}: the losses outrun it and a source must hold n_e.

n_e [cm^{-3}]	0.1 atm	0.5 atm	2 atm
pure Kr			
10^{14}	1.1^a (2.6)	2.2^a (2.0)	3.8^a (1.6)
10^{15}	3.0^s (1.8)	5.7^s (1.4)	10.1^s (1.2)
10^{16}	8.7^s (1.3)	15.4^s (1.1)	26.5^s (1.0)
Kr + 2 % F_2			
10^{14}	1.1^a (2.6)	2.2^s (1.9)	3.8^s (1.6)
10^{15}	3.0^s (1.7)	5.6^s (1.4)	9.2^s (1.1)
10^{16}	8.5^s (1.2)	13.8^s (0.2)	21.8^s (0.1)

marked ^{a} (pure Kr at 10^{14} cm^{-3}, and Kr/F_2 at 10^{14} cm^{-3} and 0.1 atm). Where $\nu_{\rm net} < 0$ a source must supply $-\nu_{\rm net} n_e$ electrons per second, at the least at the ionization energy of krypton,

$$\boxed{P_{\rm src} \geq n_e \left(\nu_{\rm att} + \nu_{\rm rec} - \nu_{\rm ion}\right) \times 14.0 \text{ eV}.} \tag{6.2}$$

In pure Kr, where recombination is the loss, that is 300 to 410 kW/cm^3 at 10^{15} and 41 to 49 MW/cm^3 at 10^{16} cm^{-3}. With 2 % F_2 the loss is attachment at 5.8×10^7 to 1.3×10^{10} s^{-1}, and the source needs 32 and 430 kW/cm^3 at 10^{14} cm^{-3} and 0.5 and 2 atm, 0.5 to 7.8 MW/cm^3 at 10^{15} and 46 to 470 MW/cm^3 at 10^{16} cm^{-3}. Neither gas holds 10^{16} cm^{-3} at these powers, and Kr/F_2 does not hold 10^{15} either.

(3) Pure Kr left to itself settles where its ionization balances recombination, $\nu_{\rm ion}(T_e(P/n_e)) = \alpha_{\rm rec}(T_e)\, n_e$, in $1/\nu_{\rm ion} = 15$ to 19 ns: at $n_e = 4.5$, 3.6 and 2.9×10^{14} cm^{-3} for 0.1, 0.5 and 2 atm ($T_e = 2.0$, 1.6 and 1.4 eV), where $f_{\rm best} = 2.1$, 3.7 and 6.1 THz. Over 100 to 500 kW/cm^3 that is 1.6 to 6×10^{14} cm^{-3} and 1.5 to 7.2 THz.

6.3 The power laws

Fitted to the nine cells of each gas at 300 kW/cm^3,

$$\boxed{f_{\rm best} \approx C \left(\frac{n_e}{10^{15}\text{ cm}^{-3}}\right)^b \left(\frac{p}{1\text{ atm}}\right)^c \left(\frac{L}{10\text{ cm}}\right)^{1/2},} \tag{6.3}$$

with $C = 7.6$ THz, $b = 0.43$, $c = 0.40$ for pure Kr within 5.1 %, and $C = 7.1$ THz, $b = 0.41$, $c = 0.37$ for Kr/F_2 within 8.6 %. From 100 to 500 kW/cm^3 the best frequency rises by at most a factor 1.16 in pure Kr, through T_e: at 10^{15} cm^{-3} and 0.5 atm it is 4.4, 7.1 and 8.7 THz for $T_e = 1$, 2 and 4 eV. In Kr/F_2 the power moves it by up to a factor 3.4, at 10^{15} cm^{-3} and 2 atm, since at 100 kW/cm^3 the F_2 losses already hold the electrons at 0.16 eV there.

6.4 Fluorine against pure krypton

(1) At a given T_e the 2 % F_2 adds at most 4.6 % to the kernel (2.11), so the frequency is the same wherever the krypton excitation takes the power: Kr/F_2 is within 2 % of pure Kr at six cells ($T_e \geq 1.2$ eV) and 9 to 18 % below it at 10^{15} cm^{-3} and 2 atm and at 10^{16} cm^{-3} and 0.5 and 2 atm,

where the F_2 losses hold the electrons in the Ramsauer minimum.

(2) What F_2 changes is the closure. Attachment removes the electrons in 0.1 to 17 ns, so a source must hold the density nearly everywhere, and from 10^{15} cm^{-3} up the source's power exceeds the field's and its electrons set T_e (Secs. 7 and 8).

In Summary: Pure Kr wants 1.1 to 3.8 THz at 10^{14} cm^{-3}, 3.0 to 10.1 at 10^{15} and 8.7 to 26.5 THz at 10^{16}, and 2 % F_2 changes the frequency only where its losses cool the electrons. The density is the source's choice wherever F_2 is present, and neither gas holds 10^{16} cm^{-3} at these powers.

7 The kinetic check

(1) The envelope takes the electrons as Maxwellian. A two-term kinetic solution [4] of every cell, with the same cross sections, the elastic collisions with the ions, electron–electron collisions in Landau's form, and the temporal-growth model in which the electrons that attachment removes or ionization adds are replaced or removed with the shape of the distribution, gives T_e as two thirds of the mean energy and the absorption from the complex mobility at the envelope's f_{best}, from which the kinetic f_A follows with $\alpha \propto f^{-2}$.

(2) In pure Kr the kinetic T_e is 0.94 to 1.35 times the envelope's and the kinetic f_{best} 0.89 to 1.23 times it: the inelastic collisions deplete the tail that a Maxwellian keeps, so the bulk must be hotter to lose the same power, most at 10^{15} cm^{-3}. With 2 % F_2 the same holds at the four cells where the field outweighs attachment, with f_{best} at 0.96 to 1.12 times the envelope. At the other five, 10^{15} cm^{-3} from 0.5 atm up and all of 10^{16}, attachment takes 0.6 to 25 times the field's power from the electrons, so the source that replaces them sets T_e, and the growth model is no longer a field-heated plasma.

In Summary: The Maxwellian envelope is good to −11 to +23 % in frequency where the field sets the temperature. Where the source does, the temperature, and with it the frequency, is the source's.

8 The uncertainty of the result

8.1 Propagation

(1) From Eq. (4.2), $f_{\text{best}} \propto \nu_{\text{eff}}^{1/2}$, so that a fractional change of the collision frequency moves the frequency by half of it, and a change of the ions' term alone by half of its share $s_i = \nu_{ei}/\nu_{\text{eff}}$:

$$\boxed{\frac{\delta f_{\text{best}}}{f_{\text{best}}} = \frac{1}{2}\,\frac{\delta \nu_{\text{eff}}}{\nu_{\text{eff}}}, \qquad \frac{\delta f_{\text{best}}}{f_{\text{best}}} = \frac{s_i}{2}\,\frac{\delta \nu_{ei}}{\nu_{ei}}.} \tag{8.1}$$

A change of T_e acts through $\sigma_m(T_e)$: between 1 and 2 eV in pure Kr, $f_{\text{best}} \propto T_e^{0.70}$, and between 2 and 4 eV, $T_e^{0.31}$.

(2) Each cell gets a central value and one uncertainty, $f_{\text{best}} \pm \delta$. With the bands $[m_-, m_+]$,

$[i_-, i_+]$, $[e_-, e_+]$ and $[x_-, x_+]$ of the four sources below as factors on the envelope's value f_{env},

$$\boxed{\begin{aligned} f_c &= f_{\text{env}}\,\frac{m_- + m_+}{2}\,\frac{i_- + i_+}{2}, \\ \frac{\delta}{f_c} &= \Big[\Big(\frac{m_+ - m_-}{2}\Big)^2 + \Big(\frac{i_+ - i_-}{2}\Big)^2 + \Big(\frac{e_+ - e_-}{2}\Big)^2 + \Big(\frac{x_+ - x_-}{2}\Big)^2\Big]^{1/2}, \end{aligned}} \tag{8.2}$$

the half-widths added in quadrature as independent errors.

8.2 The sources

(M) *The envelope against the kinetic solution.* The two models of Sec. 7 bracket the answer: $m_\mp$ are the smaller and the larger of 1 and $f_{\text{kin}}/f_{\text{env}}$. The half-width is 0.2 to 12 % in pure Kr. At the five source-set cells of Kr/F_2 the band is instead f_{best} over $T_e = 1$ to 4 eV, 0.79 to 2.5 times the envelope, and its half-width, 27 to 62 %, stands in for (M).

(I) *The electron–ion term against the measurements.* Turnbull et al. measured absorption fractions of 1 to 4 % in hydrogen and nitrogen–hydrogen plasmas with an absolute uncertainty of (−0.12 %, +0.07 %), found their best shots predicted to better than 10 %, and their models 16 % above the data on average without the fitted screening factor [10, 11]. The electron–ion absorption of Eqs. (2.13) and (2.14), with the dynamic cutoff and the exact Gaunt factor and no fitted factor, is 1.19 to 1.20 times their four hydrogen measurements. Taking the measurements as right within 10 %, the ion term is 0.75 to 0.92 of the computed one, and Eq. (8.1) with the midpoint 0.84 gives $i_\mp$: the half-width is 0.0 to 1.0 % in pure Kr, the most at 10^{16} cm^{-3} and 0.1 atm, and up to 4.3 % at the cold cells, where the ions carry most of the absorption.

(E) *The momentum-transfer cross section against swarm experiments.* The atoms carry the absorption, so the cross section σ_m of krypton is the input that matters, and the electron drift velocity W in a DC field is its experimental test: for $\omega \gg \nu$ the absorption weights σ_m as Eq. (2.11) does, and the drift velocity weights $1/\sigma_m$ over the same electron energies, so that $\delta W/W \simeq -\delta\sigma_m/\sigma_m$. The two-term equation [4] with the same Biagi set gives W at each field strength of two measurements in pure krypton, Pack, Voshall and Phelps [13] at 300 K, as compiled by Dutton [14], and Nakamura [15], read from the figure of the published abstract to about 3 %. Each measured point is placed at the electron temperature $T_e = \frac{2}{3}\langle\varepsilon\rangle$ of the computed swarm. Between 0.95 and 2.6 eV, the range of the warm cells, the computed W is 0.96 to 1.02 times Nakamura's (rms 1.8 %, 13 points) and 0.84 to 0.97 times Pack's (rms 8.4 %, 7 points), and the two experiments differ from each other by as much. Below that range only Pack's data exist, and across the Ramsauer minimum the computed W is 0.80 to 1.20 times them. For each cell the points within ±25 % of its T_e give the rms deviation of each experiment, and their quadratic mean d weights the two experiments equally. Since only the atoms' share $1 - s_i$ of ν_{eff} depends on σ_m, and $f_{\text{best}} \propto \nu_{\text{eff}}^{1/2}$, the half-width is

$$\boxed{\frac{e_+ - e_-}{2} = (1 - s_i)\,\frac{d}{2},} \tag{8.3}$$

1.4 to 4.0 % in pure Kr. The same deviation, taken as an uncertainty of σ_m, replaces an assumed one: had σ_m been taken as good to 10 %, Eq. (8.1) and the cells recomputed with $\sigma_m \times 0.9$ and $\times 1.1$ give 3.7 to 5.0 % in pure Kr. Adding both would count the cross section twice, so Eq. (8.2) carries (E), and the 10 % figure stands as a bound that the swarm data do not reach.

(X) *The inelastic cross sections of krypton*, taken as good to 20 %, move T_e through Eq. (3.1) and with it f_{best} by 1.0 to 1.7 % in pure Kr.
(D) *The recombination coefficient*, ±10 % as stated [16], moves the self-sustained density of pure Kr by −4.0 to +4.6 % and its frequency by ∓1.9 %. Its extrapolation beyond 0.72 eV is not bounded.

8.3 The result

Table 3 gives $f_{\text{best}} \pm \delta$ at the 18 cells with the four contributions. In pure Kr δ/f_{best} is 2 to 12 %, and (M) and (E) together carry most of δ^2 at eight of the nine cells. In Kr/F_2 it is 3 to 7 % at the four field-set cells and 27 to 62 % at the five source-set ones, where the temperature the source chooses is the uncertainty.

Table 3: The best frequency with its uncertainty at every cell of Table 2, and the four contributions to δ in per cent. ^{t}: the source sets T_e, and the half-width of f_{best} over $T_e = 1$ to 4 eV stands in for (M).

gas	n_e [cm^{-3}]	p [atm]	$f_{\text{best}} \pm \delta$ [THz]	δ [%]	(M)	(I)	(E)	(X)
Kr	10^{14}	0.1	1.1 ± 0.1	5	3	0.0	4.0	1.4
Kr	10^{14}	0.5	2.1 ± 0.1	5	4	0.0	2.6	1.6
Kr	10^{14}	2	3.6 ± 0.2	6	5	0.0	2.6	1.6
Kr	10^{15}	0.1	3.1 ± 0.2	6	5	0.1	2.9	1.7
Kr	10^{15}	0.5	6.2 ± 0.6	9	9	0.0	1.7	1.7
Kr	10^{15}	2	11.3 ± 1.3	12	12	0.0	1.9	1.7
Kr	10^{16}	0.1	8.5 ± 0.2	2	0	1.0	1.4	1.0
Kr	10^{16}	0.5	15.8 ± 0.6	4	3	0.4	1.6	1.4
Kr	10^{16}	2	27.9 ± 1.7	6	5	0.1	2.0	1.4
Kr/F_2	10^{14}	0.1	1.1 ± 0.1	5	2	0.0	4.0	1.4
Kr/F_2	10^{14}	0.5	2.2 ± 0.1	3	1	0.0	2.6	1.6
Kr/F_2	10^{14}	2	4.0 ± 0.2	5	5	0.0	1.9	1.5
Kr/F_2	10^{15}	0.1	3.1 ± 0.2	7	6	0.1	2.9	1.6
Kr/F_2	10^{15}	0.5	6.6 ± 2.5	38	38^t	0.0	1.7	1.5
Kr/F_2	10^{15}	2	13.1 ± 6.1	47	47^t	0.0	1.8	1.1
Kr/F_2	10^{16}	0.1	10.0 ± 2.7	27	27^t	1.1	1.4	0.7
Kr/F_2	10^{16}	0.5	19.5 ± 9.2	47	47^t	4.3	0.4	0.0
Kr/F_2	10^{16}	2	38.9 ± 24.1	62	62^t	3.7	2.1	0.0

In Summary: The uncertainty of the best frequency is Eq. (8.2) from the Maxwellian envelope against the kinetic solution, the electron–ion term against Turnbull's data, the momentum-transfer cross section against the swarm measurements, and the inelastic cross sections. It is 2 to 12 % in pure Kr, since the atoms carry the absorption, their cross section reproduces the measured drift velocities to a few per cent, and the ion term that Turnbull's measurements test contributes at most 1 %, and it is the source's temperature wherever F_2 makes a source necessary.

9 A worked example

Pure Kr at $n_e = 10^{15}$ cm^{-3}, $p = 0.5$ atm, $P = 300$ kW/cm^3 and $L = 10$ cm, the central cell of Table 2, carried through from the gas density to the recommended frequency.

(1) *The gas and the power per electron.* $N = 2.45 \times 10^{19} \times 0.5 = 1.22 \times 10^{19}$ atoms per

cm^3, and $P/n_e = 3 \times 10^5$ W cm$^{-3}/10^{15}$ cm$^{-3} = 1.87 \times 10^9$ eV/s per electron.

(2) *The temperature.* Equation (3.1) balances at $k_B T_e = 1.42$ eV: the recoil of Eq. (2.12) takes 3.3×10^7 eV/s, 1.8 %, and krypton's levels take 1.84×10^9 eV/s, the rest.

(3) *The collision frequency.* The kernel (2.11) gives $\nu_{ea} = 1.22 \times 10^{12}$ s^{-1}, 2.10 times the Drude mean 5.8×10^{11} s^{-1}. For the ions, Eq. (2.13) gives $\nu_T = 1.71 \times 10^9$ s^{-1}, $\eta_t = 4.4$ puts the collisions in the classical regime, $\Lambda = 3.65$ where the Born logarithm would give 4.90, and $\nu_{ei} = 6.3 \times 10^9$ s^{-1}, 0.5 % of $\nu_{\text{eff}} = 1.22 \times 10^{12}$ s^{-1}. So $\nu_{\text{eff}}/2\pi = 0.19$ THz.

(4) *The cutoff and the reflection limit.* $f_{pe} = 8.98$ kHz $\times (10^{15})^{1/2} = 0.28$ THz, and $f_R = 1.294 f_{pe} = 0.37$ THz without the collisions, 0.32 THz with them in Eq. (2.4).

(5) *The absorption length.* Equation (4.2) with $n_e/10^{15} = 1$, $\nu_{\text{eff}}/10^{12} = 1.22$ and $L/10$ cm $= 1$ gives $f_A = 5.2 \times (1.22)^{1/2} = 5.73$ THz, and the exact root of $\alpha L = 1$ is 5.73 THz. So $f_{\text{best}} = 5.7$ THz, far above f_R. There $u = 0.017$ and $\alpha_L = 2.2 \times 10^{-3}$, so the soft-photon kernel and the absence of Langdon's flattening both hold.

(6) *The field.* Equation (3.3) gives $E_0 = 47.6$ kV/cm, an intensity of 3 MW/cm^2.

(7) *GHz instead?* At 95 GHz, $\nu_{\text{eff}}/\omega = 2.1$: the wave is in the collisional skin, with an absorption length of 0.17 mm and 31 % reflected. At 2.45 GHz, 1.1 mm and 80 % reflected. At 1 THz the gas is already transparent, $R = 0$, but the absorption length is 3.0 mm and the front centimetre takes everything. At 3 THz it is 2.7 cm. The depth of the gas is matched on the transparent side, at 5.7 THz.

(8) *The density.* Direct ionization runs at $\nu_{\text{ion}} = 1.6 \times 10^7$ s^{-1} against recombination at 1.8×10^8 s^{-1} (5.7 ns per electron), so this density decays unless a source supplies at least 360 kW/cm^3 by Eq. (6.2), more than the field. Left to itself at this pressure and power, pure Kr settles at 3.6×10^{14} cm^{-3} and $k_B T_e = 1.62$ eV in 17 ns, where $f_{\text{best}} = 3.7$ THz.

(9) *The uncertainty.* The kinetic solution of Sec. 7 gives $k_B T_e = 1.80$ eV and $f_{\text{best}} = 6.7$ THz, 1.17 times the envelope, so (M) contributes a half-width of 8.7 %; (I) 0.02 %, since the ions carry 0.5 %; (E) 1.7 %, from the 9 swarm points within ±25 % of 1.42 eV, where the computed drift velocity deviates from Nakamura's by 1.2 % rms and from Pack's by 4.6 %, so that $d = 3.3$ %; (X) 1.7 %. Equation (8.2) gives $\delta/f_{\text{best}} = 9.0$ % about the midpoint:

$$\boxed{f_{\text{best}} = 6.2 \pm 0.6 \text{ THz.}} \tag{9.1}$$

(10) *With 2 % F_2 at the same point.* $k_B T_e = 1.37$ eV, F_2's channels taking 26 % of the losses, and $f_{\text{best}} = 5.6$ THz, 2 % below pure krypton's. But attachment runs at 6.1×10^8 s^{-1}, 1.6 ns per electron, the source must supply at least 1.7 MW/cm^3, six times the field's power, and in the kinetic solution it takes 0.60 of the field's power from the electrons, so it sets T_e and the frequency is 6.6 ± 2.5 THz, 38 %.

In Summary: At 10^{15} cm^{-3}, 0.5 atm and 300 kW/cm^3 in a 10 cm gas, the best frequency is 6.2 ± 0.6 THz, Eq. (9.1). The GHz range is out: 31 to 80 % reflected and the rest absorbed in 0.2

to 1 mm. The density is not free: pure Kr at this density needs a source of 360 kW/cm^3 or settles at 3.6×10^{14} cm^{-3}, where it wants 3.7 THz, and with F_2 the source needs 1.7 MW/cm^3 and sets the temperature.

10 Summary

In Summary: The best frequency for inverse-bremsstrahlung heating of a krypton laser gas is the one whose absorption length matches the depth of the gas, Eq. (1.1), and it is Eq. (4.2): the square root of the electron density, of the collision frequency and of the depth. The collision frequency is that of the atoms, Eq. (2.11), which the ions supplement by a few per cent at 10^{15} cm^{-3} and by up to a quarter at 10^{16}, and it depends on the electron temperature that the power balance (3.1) sets, 0.95 to 2.6 eV where the field sets it. Pure Kr wants 1.1 ± 0.1 THz at 10^{14} cm^{-3} and 0.1 atm, rising as $(n_e p)^{0.4}$ and $L^{1/2}$ to 27.9 ± 1.7 THz at 10^{16} cm^{-3} and 2 atm, with 2 to 12 % uncertainty, and left to itself it runs at 1.6 to 6×10^{14} cm^{-3} and wants 1.5 to 7 THz. The THz range is the right one: across the spectrum the absorption peaks at 0.05 to 0.50 THz, between the bands, and the GHz range either reflects, 31 to 94 % from 10^{15} cm^{-3} up, or is absorbed in a millimetre skin, while the THz range enters the gas and deposits over centimetres. 2 % F_2 leaves the frequency alone wherever the krypton excitation takes the power. But F_2 attaches the electrons in 0.1 to 17 ns, so a source sets the density and, from 10^{15} cm^{-3} up, the temperature, and the frequency is then known only to the source's temperature, 27 to 62 %. Neither gas holds 10^{16} cm^{-3} at 100 to 500 kW/cm^3. At the worked example, pure Kr at 10^{15} cm^{-3} and 0.5 atm, the answer is 6.2 ± 0.6 THz.

A The numerical checks

The script `envelope_v6_check.py` verifies every boxed result of these notes, in their order, by an independent numerical route: the Drude optics against the exact complex refractive index, the energy-weighted kernel (2.11) against the velocity-space integral (2.10) by quadrature and against the 4/3 of a constant cross section, the rate (3.2) and the recoil loss (2.12) against quadrature and against the closed form of a step cross section, the Spitzer prefactor, the Born and classical limits of the logarithm, the reflection limit (4.1), the closed form (4.2) against the exact root, the field (3.3) and Langdon's parameter (2.17) against the solved cells, the particle balance, the fits (6.3), the collisional-skin law (5.1), the swarm comparison (8.3) with the drift velocities recomputed and the measured values read from their data files, the combination (8.2), and every step of the worked example of Sec. 9. It reproduces every number printed in these notes to its last printed digit from the stored results of `envelope_v6.py` and `swarm_kr.py`, checks the quoted figures of Turnbull et al. against the text of their papers, and checks the logical statements. It runs with `numpy` and `scipy` under `/usr/bin/python3` in a few seconds from those stored results, which the scripts behind them take about four minutes to produce. Table 4 is written by the script itself. A quantitative check passes when its deviation is below the tolerance stated in the script, a printed number when the script reproduces it to its last printed digit, and a logical condition when it holds.

Table 4: The numerical checks of `envelope_v6_check.py`, by the equation, table or section they verify. The second column counts all checks of the row. The third gives the largest deviation among its quantitative checks, relative, or absolute for quantities near zero, and is empty when the row has none. The fourth counts the numbers printed in these notes that the script reproduces to their last printed digit, and the fifth the logical conditions.

Equation	Checks	Largest deviation	Printed	Logical
Sec. 1	2		1	1
Eq. (1.1)	5		2	3
Eq. (2.3)	2	0		1
Eq. (2.4)	10	1.3×10^{-9}	6	2
Eq. (2.5)	2	1.0×10^{-4}		
Eq. (2.6)	1	0		
Eq. (2.2)	1	0		
Eq. (2.11)	8	1.7×10^{-7}	5	
Sec. 2.2	1		1	
Eq. (2.12)	4	1.7×10^{-7}	2	1
Eq. (2.13)	5	1.3×10^{-3}	2	
Eq. (2.15)	3	1.1×10^{-9}	1	
Eq. (2.14)	4	4.2×10^{-3}	1	1
Sec. 2.3	7	2.7×10^{-6}	6	
Eq. (2.16)	12		9	3
Sec. 2.4	2		1	1
Eq. (2.17)	4	5.7×10^{-2}	2	1
Eq. (3.1)	6	1.3×10^{-13}	3	1
Eq. (3.2)	3	8.3×10^{-5}		
Sec. 3	21		19	2
Eq. (3.3)	6	2.7×10^{-10}	4	1
Eq. (4.1)	10	1.0×10^{-13}	7	1
Sec. 4.1	2		1	1
Eq. (4.2)	8	2.2×10^{-3}	5	2
Sec. 5	42		36	6
Eq. (5.1)	5	0	3	1
Table 1	10			10
Table 2	19			19
Eq. (6.1)	6		3	3
Sec. 6.2	36		33	3
Eq. (6.2)	4		1	3
Eq. (6.3)	9	0		3
Sec. 6.3	7		6	1
Sec. 6.4	7		5	2
Sec. 10	15		14	1
Sec. 6	1			1
Sec. 7	16		13	3
Eq. (8.1)	3	3.9×10^{-3}	2	
Eq. (8.2)	11		8	3
Sec. 8.2	29		21	8

Equation	Checks	Largest deviation	Printed	Logical
Eq. (8.3)	23	0	11	8
Sec. 8.3	7		6	1
Table 3	19			19
Sec. 9	22	0	16	4
Eq. (9.1)	5	0	2	1
Appendix A	4			4
Total	429		258	126

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